

# QpiAI™ - Joint AI Training and Certification Program

Principal Component Analysis

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Learning with the help  
of training examples  
without the use of  
training labels.

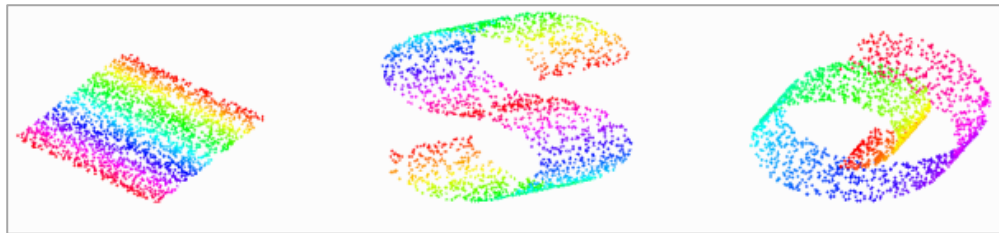
Clustering (K-Means, GMM, etc.)

Manifold Learning (PCA, Kernel PCA, t-SNE  
etc.)

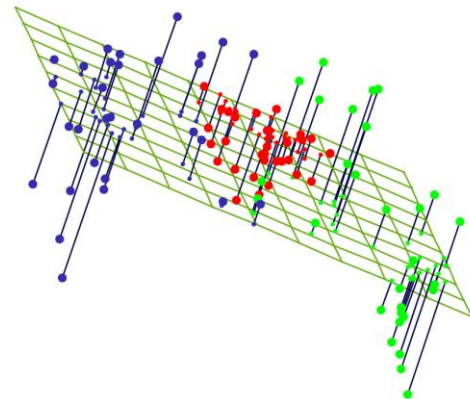
Recovering low dimensional data representation  
embedded within a higher dimensional space

Embeddings may be linear or non-linear

# Manifold learning



- Three (simple) examples of manifolds
- All three are 2D Data embedded in 3D
  - Linear, “S”-shape, “Swiss roll”
- For all three, we would like recover
  - a. A 2D representation of the data
  - b. Location of the data in 2D



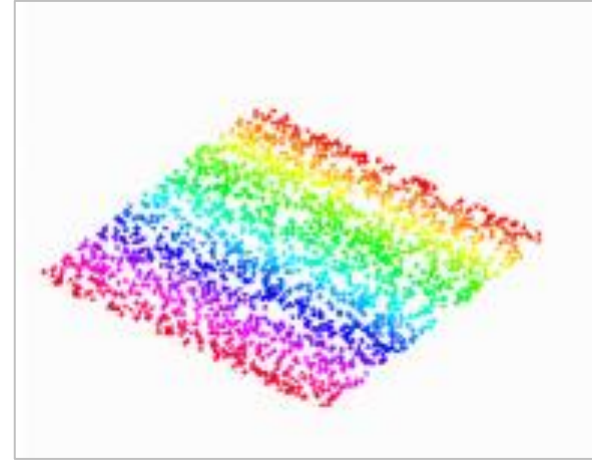
- Simple example of Linear Manifold Learning using Principal Components Analysis

# Manifold learning - applications

Image Compression

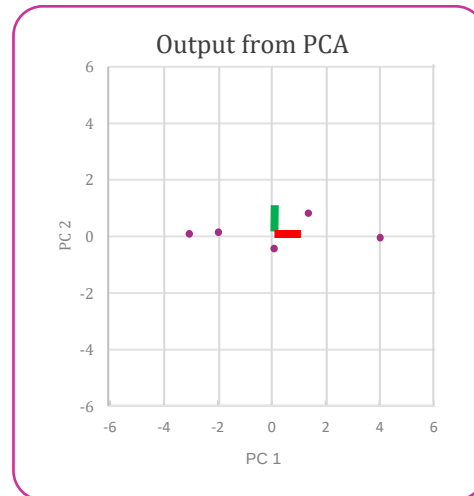
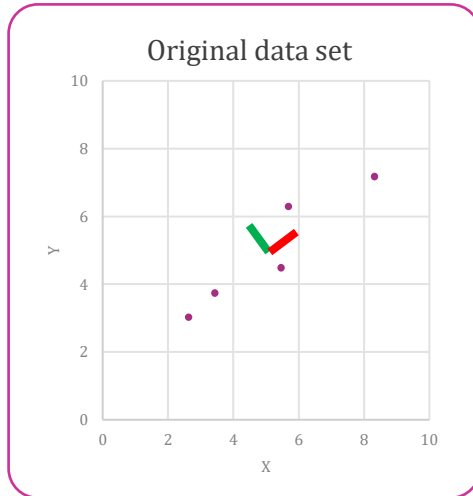


Dimensionality reduction



# Introduction to Principal Component Analysis

- Principal Component Analysis is the process of computing the principal components and using them to perform a change of basis on the data, using only the first few principal components since they capture the maximum variance within the dataset, thereby reducing the dimension of the original dataset.
- PCA produces a low-dimensional representation of a dataset. It finds a sequence of linear combinations of the variables that have maximal variance, and are mutually orthogonal (uncorrelated).
- Apart from producing derived variables for use in Supervised Learning problems, PCA also serves as a tool for data visualization.



The first principal component of a set of features  $X_1, X_2, \dots, X_p$  is the normalized linear combination of the features:

$$Z_1 = \phi_{11}X_1 + \phi_{21}X_2 + \dots + \phi_{p1}X_p$$

that has the largest variance. By normalized, we mean:

$$\sum \phi_{j1}^2 = 1$$

We refer to the elements  $\phi_{11}, \dots, \phi_{p1}$  as the elements of the first principal component; together they make up the principal component vector, the direction in feature space along which the data varies the most.

$$\phi_1 = (\phi_{11} \ \phi_{21} \ \dots \ \phi_{p1})^T$$

We constrain the elements so that their sum of squares is equal to one, since otherwise setting these elements to be arbitrarily large in absolute value could result in an arbitrarily large variance.

Suppose we have a  $n \times p$  dataset  $\mathbf{X}$ . There are at most  $\min(n - 1, p)$  principal components. Since we are interested in variance, we assume that each of the variables in  $\mathbf{X}$  has been centered to have mean zero (that is, the column means of  $\mathbf{X}$  are zero). We then look for the linear combination of the sample feature values of the form:

$$z_{i1} = \phi_{11}x_{i1} + \phi_{21}x_{i2} + \dots + \phi_{p1}x_{ip}$$

for  $i = 1, \dots, n$  that has largest sample variance, subject to the constraint:

$$\sum \phi_{j1}^2 = 1$$

Since each of the  $x_{ij}$  has mean zero, then so does  $z_{i1}$  (for any values of  $\phi_{j1}$ ). Hence the sample variance of the  $z_{i1}$  can be written as:

$$1/n \sum z_{i1}^2$$

We refer to  $Z_1$  as the first principal component, with realized values  $z_{11}, \dots, z_{n1}$ . If we project the  $n$  data points  $x_1, \dots, x_n$  onto the direction of  $\phi_1$ , the projected values are the principal component scores  $z_{11}, \dots, z_{n1}$  themselves.



## Subsequent Principal Components

- It turns out that constraining  $Z_2$  to be uncorrelated with  $Z_1$  is equivalent to constraining the direction of  $\phi_1$  to be orthogonal (perpendicular) to the direction of  $\phi_2$ . And so on, for all subsequent principal components.

$$\langle \phi_i, \phi_j \rangle = 0 \text{ for all } i \neq j$$

- The principal component directions  $\phi_1, \phi_2, \dots, \phi_p$  are the ordered sequence of right singular vectors of the matrix  $\mathbf{X}$ , and the variances of the components are  $1/n$  times the squares of the singular values.

## Linear Independence

- Vectors  $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r$  are said to be **linear dependent** if we can find  $r$  real number  $c_1, c_2, \dots, c_r$ , not all of them equal to zero, such that

$$\mathbf{0} = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_r \mathbf{v}_r$$

- Otherwise, are  $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r$  are said to be **linearly independent**
- In other words,  $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r$  are be **linearly independent** if, the only choice of  $c_1, c_2, \dots, c_r$  such that:

$$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_r \mathbf{v}_r = \mathbf{0}$$

is when  $c_1 = c_2 = \dots = c_r = 0$

## Rank of a matrix

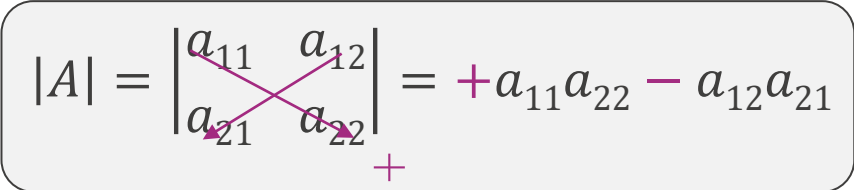
- The number of linearly independent columns or rows of matrix

## L2 norm

L2 norm of a vector  $v = [v_1, v_2, \dots, v_d] \in \mathbb{R}^d$  is given by  $\|v\|_2 = (v_1^2 + v_2^2 + \dots + v_d^2)^{1/2}$

## Determinants

Determinant of **order 2**

$$|A| = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = +a_{11}a_{22} - a_{12}a_{21}$$


**Example:** Evaluate the determinant:  $\begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix}$

$$\begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} = 1 \times 4 - 2 \times 3 = -2$$

## Determinants

Determinant of **order 3**

Consider an example:  $A = \begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix}$

Its determinant can be obtained by:

$$|A| = \begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix} = 1 \begin{vmatrix} 5 & 6 \\ 8 & 9 \end{vmatrix} - 2 \begin{vmatrix} 4 & 6 \\ 7 & 9 \end{vmatrix} + 3 \begin{vmatrix} 4 & 5 \\ 7 & 8 \end{vmatrix}$$

$$= 1(-3) - 2(-6) + 3(-3) = 0$$

## Determinants

The following properties are true for determinants of any order

- If every element of a row (column) is zero

e.g.,  $\begin{vmatrix} 1 & 2 \\ 0 & 0 \end{vmatrix} = 1 \times 0 - 2 \times 0 = 0$ , then  $|A| = 0$

- $|A^T| = |A|$                       Determinant of a matrix is equal to that of its transpose
- $|AB| = |A||B|$                 Determinant of a product of matrices is equal to the product of determinants of each matrix

## Eigenvalues & Eigenvectors

- **Eigenvectors** (for a square  $m \times m$  matrix  $S$ )

$$Sv = \lambda v$$

eigenvector  
 $v \in \mathbb{R}^m \neq 0$

eigenvalue  
 $\lambda \in \mathbb{R}$

### Example

$$\begin{pmatrix} 6 & -2 \\ 4 & 0 \end{pmatrix} \begin{pmatrix} 2 \\ 4 \end{pmatrix} = 2 \begin{pmatrix} 2 \\ 4 \end{pmatrix}$$

- How many eigenvalues are there at most?

$$Sv = \lambda v \Leftrightarrow (S - \lambda I)v = 0$$

only has a non-zero solution if  $|S - \lambda I| = 0$  this is  $m^{\text{th}}$  order equation in  $\lambda$  which can have **at most  $m$  distinct solution** (roots of the characteristic polynomial) – can be complex even though  $S$  is real

## Eigenvalue Eigenvector example

- Task: find eigenvalues of  $A = \begin{pmatrix} 3 & -1 & -1 \\ -12 & 0 & 5 \\ 4 & -2 & -1 \end{pmatrix}$

- Characteristic polynomial is  $\begin{vmatrix} 3-\lambda & -1 & -1 \\ -12 & -\lambda & 5 \\ 4 & -2 & -1-\lambda \end{vmatrix}$

$$= (3 - \lambda) \begin{vmatrix} -\lambda & 5 \\ -2 & -1 - \lambda \end{vmatrix} + 1 \begin{vmatrix} -12 & 5 \\ 4 & -1 - \lambda \end{vmatrix} - 1 \begin{vmatrix} -12 & -\lambda \\ 4 & -2 \end{vmatrix}$$

$$= (3 - \lambda)(\lambda(1 + \lambda) + 10) + 1(12(1 + \lambda) - 20) - 1(24 + 4\lambda)$$

$$= (3 - \lambda)(\lambda^2 + \lambda + 10) + 12\lambda - 8 - 24 - 4\lambda$$

$$= 3\lambda^2 + 3\lambda + 30 - \lambda^3 - \lambda^2 - 10\lambda + 12\lambda - 8 - 24 - 4\lambda$$

$$= -\lambda^3 + 2\lambda^2 + \lambda - 2$$

# Eigenvalue Eigenvector example

- We need to **solve**  $-\lambda^3 + 2\lambda^2 + \lambda - 2 = 0$
- We try a few “obvious” values:  **$\lambda=1$**  YES!
- Reduce from degree 3 to 2, by separating  $(\lambda-1)$  in:  

$$-\lambda^3 + 2\lambda^2 + \lambda - 2 = (\lambda - 1)(a\lambda^2 + b\lambda + c)$$

$$= a\lambda^3 + (b - a)\lambda^2 + (c - b)\lambda - c$$
- This works for  $a = -1, b = 1, c = 2$
- Alternatively, we can factor out  $(\lambda - 1)$  to get the values of a, b and c as:  $-\lambda^2 + \lambda + 2 = 0$
- Solution:  $\lambda = \frac{-1 \pm \sqrt{1+4*2}}{-2} = \frac{-1 \pm 3}{-2}$  giving  $\lambda=2, -1$
- All three eigenvalues:  **$\lambda = 1, \lambda = -1, \lambda = 2$**



## Eigenvalue Eigenvector example

- Once we have eigenvalues  $\lambda_i$  for a matrix  $A$  we can find corresponding **eigenvectors**  $\mathbf{v}_i$ , with  $\mathbf{A} \cdot \mathbf{v}_i = \lambda_i \mathbf{v}_i$
- These  $\mathbf{v}_i$  appear as the solutions of  $(\mathbf{A} - \lambda_i \mathbf{I}) \cdot \mathbf{v} = 0$ 
  - We can make a convenient choice, using that scalar multiplications  $a \cdot \mathbf{v}_i$  are also a solution
- We use standard techniques for solving such equations

## Eigenvalue Eigenvector example

- Recall the eigenvalues  $\lambda = -2, \lambda = 6$  for  $A = \begin{pmatrix} 1 & 5 \\ 3 & 3 \end{pmatrix}$

$\lambda = -2$  Gives matrix  $A - \lambda I = \begin{pmatrix} 1+2 & 5 \\ 3 & 3+2 \end{pmatrix} = \begin{pmatrix} 3 & 5 \\ 3 & 5 \end{pmatrix}$

Corresponding system of equation  $\begin{cases} 3x + 5y = 0 \\ 3x + 5y = 0 \end{cases}$

- Solution choice  $x=-5, y=3$ , so  $(-5,3)$  is **eigenvector** (of matrix A with eigenvalue  $\lambda=-2$ )
- Check

$$\begin{pmatrix} 1 & 5 \\ 3 & 3 \end{pmatrix} \cdot \begin{pmatrix} -5 \\ 3 \end{pmatrix} = \begin{pmatrix} -5 & +15 \\ -15 & +9 \end{pmatrix} = \begin{pmatrix} 10 \\ -6 \end{pmatrix} = -2 \begin{pmatrix} -5 \\ 3 \end{pmatrix} \quad \checkmark$$

## Eigenvalue Eigenvector example

$$\lambda = 6$$

Gives matrix  $A - \lambda I = \begin{pmatrix} 1+6 & 5 \\ 3 & 3+6 \end{pmatrix} = \begin{pmatrix} -5 & 5 \\ 3 & -3 \end{pmatrix}$

Corresponding system of equation  $\begin{cases} -5x + 5y = 0 \\ 3x - 3y = 0 \end{cases}$

- Solution choice  $x=1, y=1$ , so  $(1,1)$  is **eigenvector** (of matrix  $A$  with eigenvalue  $\lambda=6$ )
- Check

$$\begin{pmatrix} 1 & 5 \\ 3 & 3 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 & +5 \\ 3 & +3 \end{pmatrix} = \begin{pmatrix} 6 \\ 6 \end{pmatrix} = 6 \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \checkmark$$

# Eigen Decomposition/ Diagonalization

- Let  $\mathbf{S} \in \mathbb{R}^{m \times m}$   
be a square matrix with  $m$  linearly independent eigenvectors (a "non-defective" matrix)
- Theorem: There exists an **eigen decomposition**:

$$\mathbf{S} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^{-1}$$

Diagonal

Unique for distinct  
eigenvalues

(cf. matrix diagonalization theorem)

- Columns of  $\mathbf{U}$  are **eigenvectors** of  $\mathbf{S}$
- Diagonal elements of  $\mathbf{\Lambda}$  are **eigenvalues** of  $\mathbf{S}$

$$\mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_m), \lambda_i \geq \lambda_{i+1}$$

# Symmetric Eigen Decomposition

- Let  $\mathbf{S} \in \mathbb{R}^{m \times m}$  is a **symmetric** matrix:
- Theorem: Exists a (unique) **eigen decomposition**  $\mathbf{S} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^T$
- Where  $\mathbf{Q}$  is **orthogonal**:
- $\mathbf{Q}^{-1} = \mathbf{Q}^T$
- Columns of  $\mathbf{Q}$  are normalized eigenvectors
- Columns are orthogonal and values are real

# Drawbacks of Eigen Decomposition

1

Doesn't work for rectangular matrices

2

May not even exist for some square matrices

# Covariance Matrix

If there are  $N$  Random Variables  $X_1, X_2, \dots, X_N$  we study the relationships between them with the help of an  $N \times N$  Covariance Matrix.

Let  $\mathbf{X} = (X_1, X_2, \dots, X_N)^T$  Then, Covariance Matrix  $K_{\mathbf{XX}} = E[(\mathbf{X} - E(\mathbf{X}))^T (\mathbf{X} - E(\mathbf{X}))]$

$$K_{\mathbf{XX}}(i, i) = \text{Var}(X_i)$$

$$K_{\mathbf{XX}}(i, j) = \text{Cov}(X_i, X_j) \text{ for } i \neq j$$

$$K_{\mathbf{XX}} = \begin{bmatrix} E[(X_1 - E[X_1])(X_1 - E[X_1])] & E[(X_1 - E[X_1])(X_2 - E[X_2])] & \cdots & E[(X_1 - E[X_1])(X_n - E[X_n])] \\ E[(X_2 - E[X_2])(X_1 - E[X_1])] & E[(X_2 - E[X_2])(X_2 - E[X_2])] & \cdots & E[(X_2 - E[X_2])(X_n - E[X_n])] \\ \vdots & \vdots & \ddots & \vdots \\ E[(X_n - E[X_n])(X_1 - E[X_1])] & E[(X_n - E[X_n])(X_2 - E[X_2])] & \cdots & E[(X_n - E[X_n])(X_n - E[X_n])] \end{bmatrix}$$

# Singular Value Decomposition

For an  $m \times n$  matrix  $A$  of rank  $r$  there exists a factorization named as Singular Value Decomposition (**SVD**) as follows:

The diagram shows the SVD factorization  $A = U\Sigma V^T$  in a light gray rounded rectangle. Below it are three teal boxes representing dimensions:  $m \times m$  under  $U$ ,  $m \times n$  under  $\Sigma$ , and  $n \times n$  under  $V^T$ . Arrows point from each of these boxes to the corresponding matrix in the equation above.

$$A = U \Sigma V^T$$

$m \times m$        $m \times n$        $n \times n$

The columns of  $\mathbf{U}$  are orthogonal eigenvectors of  $\mathbf{A}\mathbf{A}^T$ .

The columns of  $\mathbf{V}$  are orthogonal eigenvectors of  $\mathbf{A}^T\mathbf{A}$ .

Eigenvalues  $\lambda_1, \dots, \lambda_r$  of  $\mathbf{A}\mathbf{A}^T$  are the eigenvalues of  $\mathbf{A}^T\mathbf{A}$ .

$$\sigma_i = \sqrt{\lambda_i}$$

$$\Sigma = \text{diag}(\sigma_1 \dots \sigma_r)$$

Singular values  
arranged in  
descending order



# Singular Value Decomposition

- Illustration of SVD dimensions and sparseness

$$\begin{array}{c}
 \underbrace{\begin{bmatrix} * & * & * \\ * & * & * \\ * & * & * \\ * & * & * \end{bmatrix}}_A = \underbrace{\begin{bmatrix} * & * & * & * & * \\ * & * & * & * & * \\ * & * & * & * & * \\ * & * & * & * & * \\ * & * & * & * & * \end{bmatrix}}_U \underbrace{\begin{bmatrix} \blacksquare & & & \\ & \blacksquare & & \\ & & \blacksquare & \\ & & & \blacksquare \end{bmatrix}}_\Sigma \underbrace{\begin{bmatrix} * & * & * \\ * & * & * \\ * & * & * \end{bmatrix}}_{V^T}
 \end{array}$$
  

$$\begin{array}{c}
 \underbrace{\begin{bmatrix} * & * & * & * \\ * & * & * & * \\ * & * & * & * \end{bmatrix}}_A = \underbrace{\begin{bmatrix} * & * & * \\ * & * & * \\ * & * & * \end{bmatrix}}_U \underbrace{\begin{bmatrix} \blacksquare & & & \\ & \blacksquare & & \\ & & \blacksquare & \\ & & & \blacksquare \end{bmatrix}}_\Sigma \underbrace{\begin{bmatrix} * & * & * & * & * \\ * & * & * & * & * \\ * & * & * & * & * \\ * & * & * & * & * \\ * & * & * & * & * \end{bmatrix}}_{V^T}
 \end{array}$$

## SVD example

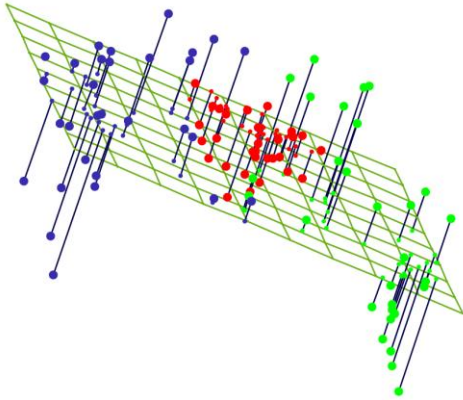
$$\text{Let } A = \begin{bmatrix} 1 & -1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Thus  $m=3$ ,  $n=2$ . Its SVD is:

$$\begin{bmatrix} \sqrt{6}/3 & 0 & -1/\sqrt{3} \\ -1/\sqrt{6} & 1/\sqrt{2} & -1/\sqrt{3} \\ 1/\sqrt{6} & 1/\sqrt{2} & 1/\sqrt{3} \end{bmatrix} \begin{bmatrix} \sqrt{3} & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$$

Typically, the singular values are arranged in decreasing order

# Principal Component Analysis (PCA)



- Linear manifold learning
- Recover a lower dimensional linear subspace embedded in higher dimensional space
- Given data  $x_1, x_2, \dots, x_n \in \mathbb{R}^d$
- Find orthogonal projection matrix  $\mathbf{U} \in \mathbb{R}^{d \times k}, (k \leq d)$
- Find principal components  $y$  by projecting data i.e.,  $y_i = \mathbf{U}^T x_i \forall i = 1, 2, \dots, n$
- Reconstruction error  $\|x - \mathbf{U}y\|_2^2 = \|x - \mathbf{U}\mathbf{U}^T x\|_2^2$  is minimized

## Method 1: PCA based on Eigen Decomposition

Input: Training examples  $x_1, x_2, \dots, x_n \in \mathbb{R}^d$ , target dimension  $k$

1. Construct data matrix  $\mathbf{X} = [x_1, x_2, \dots, x_n] \in \mathbb{R}^{d \times n}$
2. Find the mean vector  $\mu = \frac{1}{n} \mathbf{X} \mathbf{J}$  where  $\mathbf{J} = [1, 1, \dots, 1] \in \mathbb{R}^n$
3. Construct mean-centred Covariance Matrix  $\Sigma = (\mathbf{X} - \mu \mathbf{J}^T)(\mathbf{X} - \mu \mathbf{J}^T)^T \in \mathbb{R}^{d \times d}$
4. Perform Eigen decomposition of  $\Sigma$ : Find  $u_i$  such that  $\Sigma u_i = \lambda_i u_i \forall i = 1, 2, \dots, d$
5. Stack eigenvectors corresponding to top  $k$  eigenvalues in projection matrix  $\mathbf{U} = [u_1, u_2, \dots, u_k] \in \mathbb{R}^{d \times k}$
6. Find principle components matrix  $\mathbf{Y} = \mathbf{U}^T (\mathbf{X} - \mu \mathbf{J}^T) \in \mathbb{R}^{k \times n}$

## Method 2: PCA based on Singular Value Decomposition

Input: Training examples  $x_1, x_2, \dots, x_n \in \mathbb{R}^d$ , target dimension  $k$

1. Construct data matrix  $X = [x_1, x_2, \dots, x_n] \in \mathbb{R}^{d \times n}$
2. Find the mean vector  $\mu = \frac{1}{n} XJ$  where  $J = [1, 1, \dots, 1] \in \mathbb{R}^n$
3. Mean-center the data matrix  $\tilde{X} = X - \mu J^T$
4. Perform Singular value decomposition on  $\tilde{X}$ :  $\tilde{X} = U_x \Sigma_x V_x^T$
5. Stack left singular vectors in  $U_x$  corresponding to top  $k$  singular values in projection matrix  $U = [u_1, u_2, \dots, u_k] \in \mathbb{R}^{d \times k}$
6. Find principle components matrix  $Y = U^T(\tilde{X} - \mu J^T) \in \mathbb{R}^{k \times n}$

# Principal Component Analysis

Let  $x_1, x_2, \dots, x_n \in \mathbb{R}^d$  be data vectors generated from a multivariate distribution. Let  $\mu \in \mathbb{R}^d$  be the mean of each dimension in the vector space. Then, PCA can be expressed as the following optimization problem.

$$\underset{y \in \mathbb{R}^k, U \in \mathbb{R}^{d \times k}, \mu \in \mathbb{R}^d}{\text{minimize}} \sum_{i=1}^n \|x_i - \mu - Uy_i\|_2^2 \quad (1)$$

$$\text{Subject to } U^T U = I_k, \frac{1}{n} \sum_{i=1}^n y_i = 0 \quad (2)$$

Solution for  $\mu$  is given by the data mean  $\mu = \frac{1}{n} \sum_{i=1}^n x_i$

Solution for the principal components  $y_i$  is given by  $y_i = U^T (x_i - \mu)$

Substitution  $y_i$  in the original optimization problem we get

$$\underset{U \in \mathbb{R}^{d \times k}}{\text{minimize}} \sum_{i=1}^n \|x_i - \mu - UU^T(x_i - \mu)\|_2^2 \quad (3)$$

$$\text{Subject to } U^T U = I_k \quad (4)$$

# Principal Component Analysis

The above is equivalent to solving the following optimization problem

$$\max_{\mathbf{U} \in \mathbb{R}^{d \times k}} \sum_{i=1}^n (\mathbf{x}_i - \boldsymbol{\mu})^T \mathbf{U} \mathbf{U}^T (\mathbf{x}_i - \boldsymbol{\mu}) \quad (5)$$

$$\text{Subject to } \mathbf{U}^T \mathbf{U} = \mathbf{I}_k \quad (6)$$

One solution for the projection matrix  $\mathbf{U}$  is given by eigenvectors of the mean-centered covariance matrix of the data

## Image compression

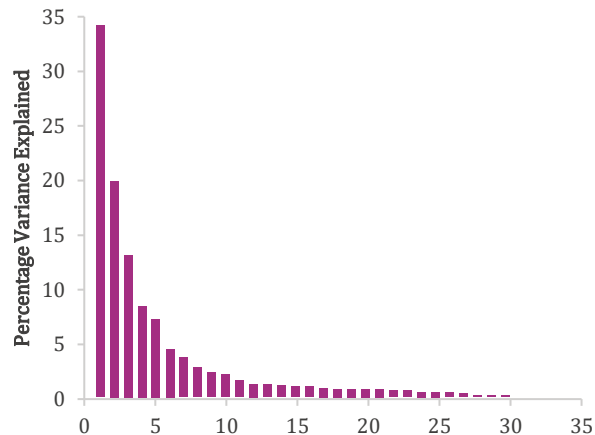
In this example, PCA is applied in the compression of 512-by-512 grey-scale image

The image represented by matrix  $x \in \mathbb{R}^{512 \times 512}$

Original greyscale image

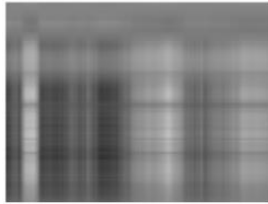


Ordered Principal Components





# PCA applications



(a) 1 principal component



(b) 5 principal component



(c) 9 principal component



(d) 13 principal component



(e) 17 principal component



(f) 21 principal component



(g) 25 principal component

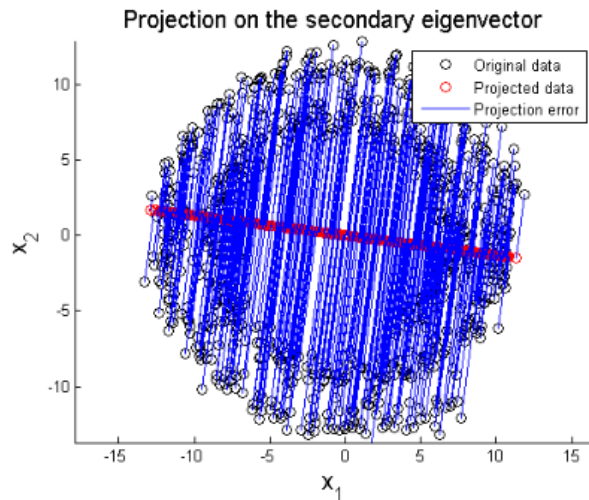
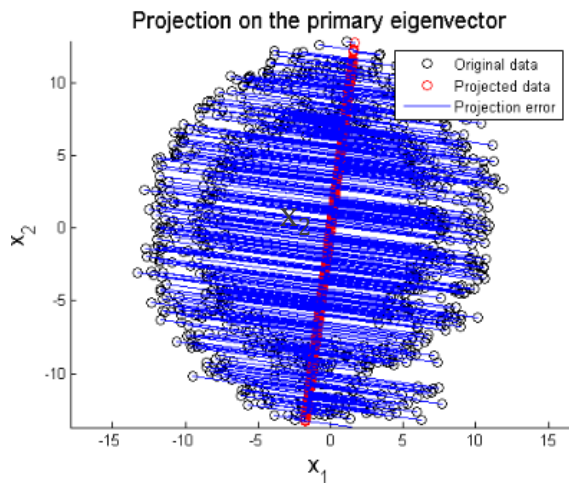
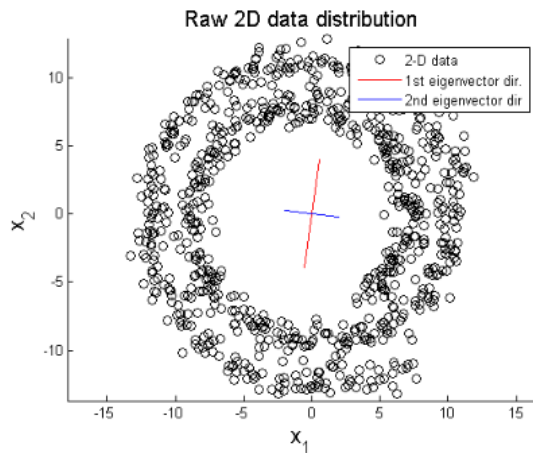


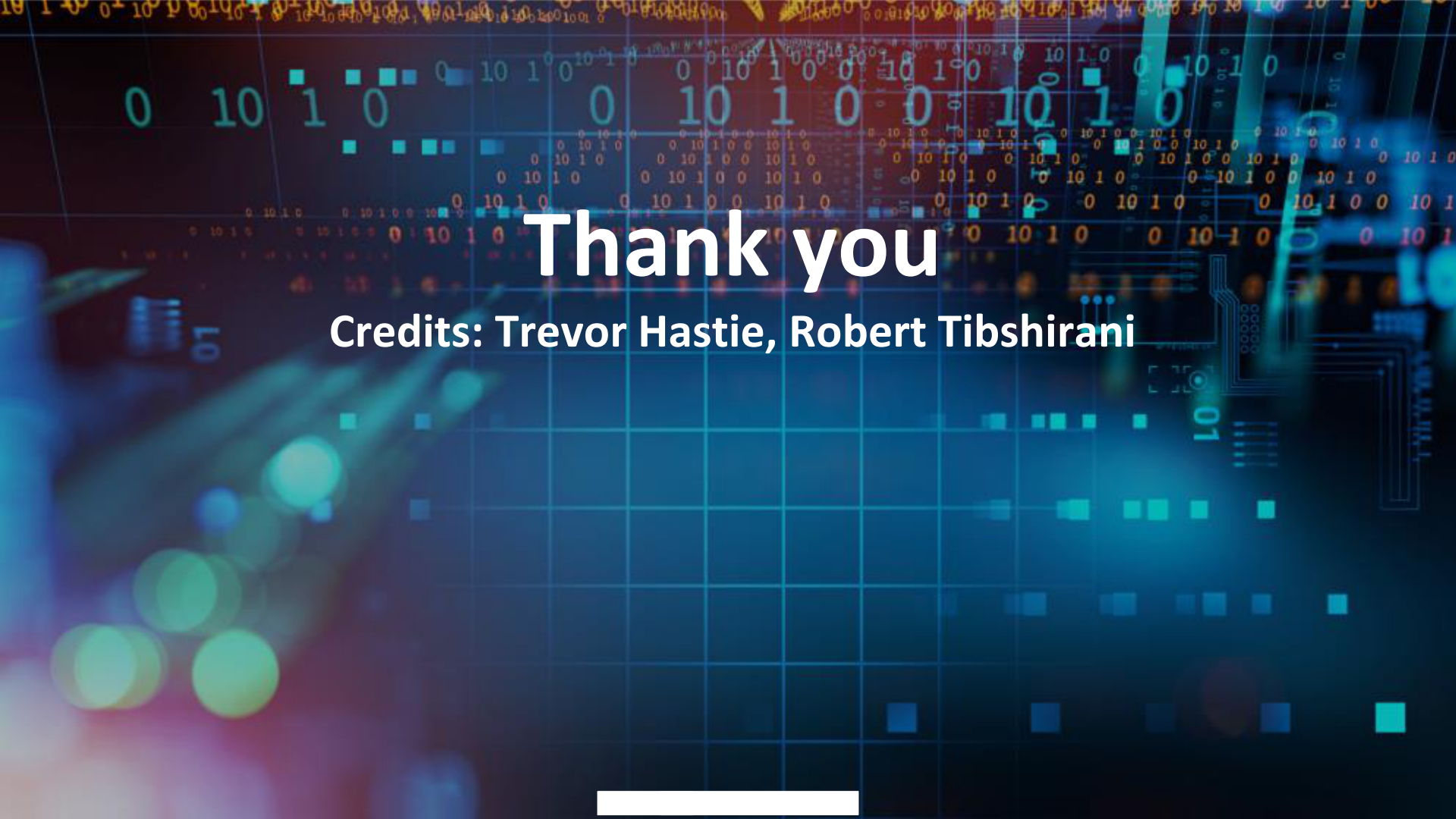
(h) 29 principal component

Figure 7. PCA results

# Drawbacks of PCA

- Doesn't work well for non-linearly distributed data or non-linear structures





# Thank you

Credits: Trevor Hastie, Robert Tibshirani